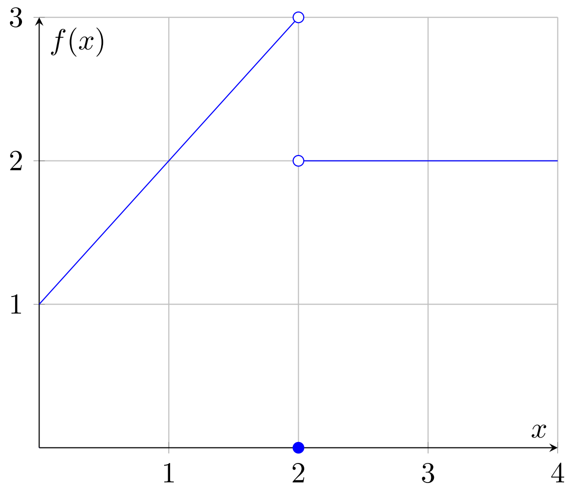
Business Calculus (MATH 2350)
Summer 2019
Friday, 05/31/2019
Quiz 1: Section 10.1, 10.3, 10.4

Name (Print): ____________________

Score:	/ 100

Section 10.1

1. Use the graph of the function f shown to estimate the indicated limits.



(a) $\boxed{\text{5 points}}$ $\lim\limits_{x\to 2^-} f(x)$

$\lim\limits_{x\to 2^-} f(x) = 3$

(b) $\boxed{\text{5 points}}$ $\lim\limits_{x\to 2} f(x)$

The limit does not exist, as $\lim\limits_{x\to 2^-} f(x) = 3$ while $\lim\limits_{x\to 2^+} f(x) = 2$

2. $\boxed{\text{10 points}}$ Find $\displaystyle\lim_{x\to 3}\frac{x-3}{x^2-9}$ or explain why the limit does not exist.

If we try to calculate the limit directly, we yield

$$\lim_{x\to 3}\frac{x-3}{x^2-9}=\frac{3-3}{(3)^2-9}\text{''}=\text{''}\frac{0}{0},$$

which is an indeterminate form. So, we must factor and cancel to yield the correct solution:

$$\lim_{x\to 3}\frac{x-3}{x^2-9}=\lim_{x\to 3}\frac{(x-3)}{(x-3)(x+3)}=\lim_{x\to 3}\frac{1}{x+3}=\frac{1}{3+3}=\frac{1}{6}.$$

3. $\boxed{\text{10 points}}$ If $f(x)=\begin{cases}x+2 & \text{if } x\leq 0\\ 2-x & \text{if } x>0\end{cases}$, find $\displaystyle\lim_{x\to 0}f(x)$.

For the limit to exist, it must match up with the left-hand and right-hand limits. First, we calculate the left-hand limit:

$$\lim_{x\to 0^-}f(x)=\lim_{x\to 0^-}x+2=0+2=2.$$

Then, we calculate the right-hand limit:

$$\lim_{x\to 0^+}f(x)=\lim_{x\to 0^+}2-x=2-0=2.$$

As these are equal, we then yield that

$$\lim_{x\to 0}f(x)=2.$$

4. Given that $\lim_{x\to 2} f(x) = 1$ and $\lim_{x\to 2} g(x) = -11$, find the indicated limits.

   (a) $\boxed{\text{5 points}}$ $\lim_{x\to 2} 2g(x) + 5f(x)$

$$\begin{aligned}\lim_{x\to 2} 2g(x) + 5f(x) &= \lim_{x\to 2} 2g(x) + \lim_{x\to 2} 5f(x)\\ &= 2\lim_{x\to 2} g(x) + 5\lim_{x\to 2} f(x)\\ &= 2(-11) + 5(1)\\ &= -17\end{aligned}$$

   (b) $\boxed{\text{5 points}}$ $\lim_{x\to 2} \frac{2x - f(x)}{x^2 + 11}$

$$\begin{aligned}\lim_{x\to 2} \frac{2x - f(x)}{x^2+11} &= \frac{\lim_{x\to 2}(2x - f(x))}{\lim_{x\to 2}(x^2+11)}\\ &= \frac{\lim_{x\to 2} 2x - \lim_{x\to 2} f(x)}{\lim_{x\to 2} x^2 + \lim_{x\to 2} 11}\\ &= \frac{4-1}{4+11}\\ &= \frac{3}{15}\\ &= \frac{1}{5}\end{aligned}$$

5. A cell phone company charges \$40.00 for the first 1000 minutes of calls per month, and \$0.15 for each additional minute.

   (a) 5 points (bonus) Write a piecewise definition of the charge $F(x)$ for x minutes of calls in one month.

$$F(x) = \begin{cases} 40 & x \leq 1000 \\ 40 + 0.15(x - 1000) & x > 1000 \end{cases}$$

i.e. \$40.00 if you use less than 1000 minutes, then \$30.00 and an additional \$0.15 for every minute beyond 1000 minute you go beyond 1000 minutes (that is,\$0.15 times $x - 1000$ minutes).

   (b) 5 points (bonus) Find $\lim_{x \to 1000} F(x)$ or explain why it does not exist. Show your work!

For the limit to exist, it must match up with the left-hand and right-hand limits. First, we calculate the left-hand limit:

$$\lim_{x \to 1000^-} F(x) = \lim_{x \to 1000^-} 40 = 40.$$

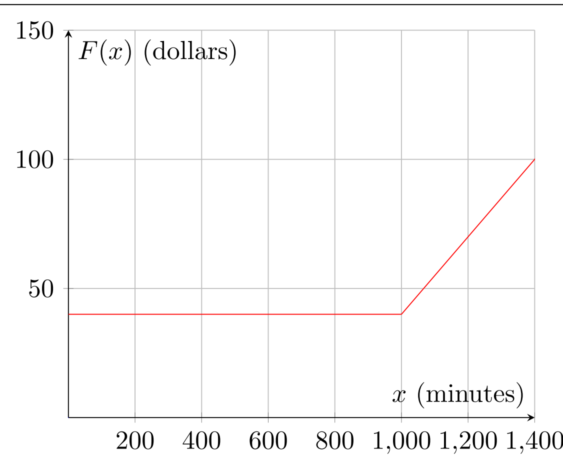
Then, we calculate the right-hand limit:

$$\lim_{x \to 1000^+} F(x) = \lim_{x \to 1000^+} 40 + 0.15(x - 1000) = 40 + 0.15(1000 - 1000) = 40 + 0.15(0) = 40.$$

As these are equal, we then yield that

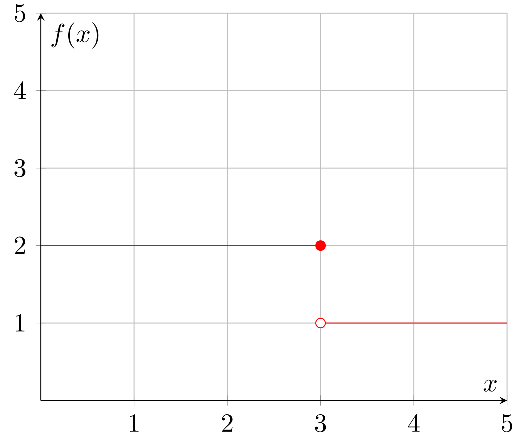
$$\lim_{x \to 1000} F(x) = 40.$$

   (c) 10 points (bonus) Graph $F(x)$. Label your graph, indicating the values of x and $F(x)$.



Section 10.3

1. $\boxed{\text{10 points}}$ Sketch a possible graph of a function f such that $\lim_{x \to 3^-} f(x) = 2$, $f(3) = 2$, and f is not continuous at $x = 3$.



2. Determine where each function is continuous. Express your answer in interval notation.

(a) $\boxed{\text{5 points}}$ $f(x) = 6x^4 + x$

A polynomial function is continuous everywhere, so $f(x) = 6x^4 + x$ is continuous on the interval $(-\infty, \infty)$.

(b) $\boxed{\text{5 points}}$ $f(x) = |x|$

The function $f(x) = |x|$ is equal to x for $x \geq 0$ and $-x$ for $x \leq 0$. As these functions are both equal to zero at $x = 0$ and are continuous (in fact, polynomial) functions elsewhere, the function $f(x) = |x|$ is continuous.

3. $\boxed{\text{10 points}}$ Let $f(x) = \begin{cases} \dfrac{x^2+x-6}{x-2} & \text{if } x \neq 2 \\ 7 & \text{if } x = 2 \end{cases}$. Is $f(x)$ continuous at $x = 2$? Explain.

For a function to be continuous at a point, the limit must exist at that point, and it must equal the value of the function at that point. If the limit exists at a point, it must match up with the left-hand and right-hand limits at that point. First, we calculate the left-hand limit:

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} \frac{x^2+x-6}{x-2} = \lim_{x\to 2^-} \frac{(x-2)(x+3)}{x-2} = \lim_{x\to 2^-} x+3 = 5.$$

Then, we calculate the right-hand limit:

$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} \frac{x^2+x-6}{x-2} = \lim_{x\to 2^+} \frac{(x-2)(x+3)}{x-2} = \lim_{x\to 2^+} x+3 = 5.$$

As these are equal, we then yield that

$$\lim_{x\to 2} f(x) = 5.$$

However, $f(2) = 7$, so while the limit does exist at $x = 2$, the limit does not equal the value of the function at $x = 2$. Thus, $f(x)$ is not continuous at $x = 2$.

Section 10.4

1. Find the indicated quantities for $f(x) = 8x^2 + 1$.

   (a) $\boxed{\text{5 points}}$ $\dfrac{f(2) - f(1)}{2 - 1}$

   $$\frac{f(2) - f(1)}{2 - 1} = \frac{(8(2)^2 + 1) - (8(1)^2 + 1)}{2 - 1} = \frac{33 - 9}{1} = 24.$$

   (b) $\boxed{\text{5 points}}$ $f'(1)$

   First, we calculate $f'(x)$:

   $$f'(x) = 8(2)x^{2-1} + 0 = 16x.$$

   Then, we calculate $f'(1)$:

   $$f'(1) = 16(1) = 16.$$

2. The revenue (in dollars) from the sale of x tables is given by

$$R(x) = 100x - 0.1x^2, \qquad 0 \leq x \leq 1000.$$

(a) $\boxed{\text{5 points}}$ Find the average change in revenue per table if production is changed from 200 to 225 tables.

$$\frac{R(225) - R(200)}{225 - 200} = \frac{100(225) - 0.1(225)^2 - 100(200) + 0.1(200)^2}{25} = 57.5.$$

(b) $\boxed{\text{10 points}}$ Find $R'(x)$ *by using the definition of the derivative* (i.e. by using limits, not the linearity of the derivative and the power rule).

$$\begin{aligned} R'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} &= \lim_{h\to 0} \frac{100(x+h) - 0.1(x+h)^2 - 100x + 0.1(x)^2}{h} \\ &= \lim_{h\to 0} \frac{100x + 100h - 0.1x^2 - 0.2xh - 0.1h^2 - 100x + 0.1x^2}{h} \\ &= \lim_{h\to 0} \frac{100h - 0.2xh - 0.1h^2}{h} \\ &= \lim_{h\to 0} \frac{h(100 - 0.2x - 0.1h)}{h} \\ &= \lim_{h\to 0} 100 - 0.2x - 0.1h \\ &= 100 - 0.2x - 0.1(0) \\ &= 100 - 0.2x \end{aligned}$$

(c) $\boxed{\text{5 points}}$ Find the instantaneous rate of change of revenue at a production level of 200 tables.

$R'(200) = 100 - 0.2(200) = 100 - 40 = 60.$